VAISHNAVI MORU

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OBJECTIVE

Analytical and highly driven Electronics and Communication Engineering student seeking a challenging role as a VLSI, RTL, or Physical Design Engineer to leverage strong skills in hardware-software co-design, logic optimization, and circuit design.

EDUCATION

B.Tech in Electronics and Communication Engineering (ECE) — Final Year (Pursuing)
University College of Engineering and Technology for Women, Warangal

Diploma in Electronics and Communication Engineering (ECE) — Graduated 2024 | CGPA: 9.5/10
Govt. Polytechnic Medak

Secondary School Certificate (10th Standard) — Completed 2021 | CGPA: 10/10
Zilla Parishad High School, Mancherial

TECHNICAL & VLSI SKILLS

Hardware Description & Verification:	Verilog HDL, SystemVerilog, UVM
VLSI & ASIC Design Flow:	RTL Design, ASIC, FPGA, Physical Design, Synthesis, Placement, Clock Tree Synthesis (CTS), Routing, Design Rule Check (DRC), Design for Testability (DFT), Power Analysis
Timing & Analysis:	Static Timing Analysis (STA), Clock Domain Crossing (CDC)
Custom IC & Layout:	CMOS Digital Design, Custom IC Design, Custom Layout using Magic tool
EDA Tools & Software:	Xilinx Vivado, ModelSim
Programming & Scripting:	C, C++, Python

ACADEMIC & VLSI PROJECTS

Traffic Light Controller & Digital Clock

- Designed a smart traffic light controller utilizing a Finite State Machine (FSM) for state transitions and clock division.
- Built a real-time digital clock in Verilog, integrating counters, clock management, and a 7-segment display interface.

Sensor Automation and Detection

- Designed and configured automated sensor arrays for real-time monitoring and event detection.
- Integrated microcontrollers and firmware to process sensor signals and trigger automated control actions efficiently.

Ultrasonic Sensor Detection

- Configured ultrasonic distance measurement modules with microcontrollers to achieve real-time object ranging and presence detection.
- Processed echo time signals to trigger alerts and control automated threshold responses.

WORK EXPERIENCE

Technical Specialist (Hardware & Firmware Validation) — July 2023 – September 2023
Glow LED

- Executed structural hardware testing and software diagnostics on micro-controlled circuit layouts, successfully isolating critical processing bottlenecks.
- Handled multi-meter continuous diagnostic operations, systematically creating defect data tables to cross-validate systems against low-level engineering layouts.

CERTIFICATIONS & PARTICIPATION

- **Microsoft:** Certificate of Completion - Introduction to Generative AI Concepts (May 2025)
- **Scaler Masterclass:** Certificate of Participation - Low-Level Design of Payment Apps (March 2026)
- **Ministry of Youth Affairs & Sports (MYBharat):** Online Quiz on MY Bharat Budget Quest 2026 (February 2026)
- **Ministry of Youth Affairs & Sports (MYBharat):** Online Quiz on Viksit Bharat Young Leaders Dialogue VBYLD 2026 (October 2025)
- **PGDCA:** Post Graduate Diploma in Computer Applications

LANGUAGES

English, Telugu